

openGUTS Report

Project:

GUTS_RF_Daphnia_81_Noprefitbackgroundhazard_07052026

Project file:

No project file saved or loaded

Project description (optional):

No project description available

Software version:

openGUTS - 1.2

Date of report creation:

07/05/2026 10:12:53

Calibration

Calibration input data

Data set 1

File: Data edited manually

Description (optional):

Control group: 'Control'

Survival data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6
0	79	19	20	29	20	30	19
0.042	79	19	19	29	20	30	19
0.083	79	19	19	29	20	30	19
0.16	79	19	19	29	20	30	19
0.33	79	19	19	29	20	30	19
1	78	19	19	29	20	30	17
1.208	78	19	19	29	20	28	16
2	78	19	19	29	17	13	6

Concentration data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6
0	0	0.3	1	3	10	30	100
0.2	0	0.3	0.99	2.97	9.9	29.7	99
0.4	0	0.29	0.98	2.94	9.79	29.4	97.9
0.6	0	0.29	0.97	2.91	9.69	29.1	96.9
0.8	0	0.29	0.96	2.88	9.59	28.8	95.9
1	0	0.28	0.95	2.85	9.49	28.5	94.9
1.2	0	0.28	0.94	2.82	9.39	28.2	93.9
1.4	0	0.28	0.93	2.79	9.29	27.9	92.9
1.6	0	0.28	0.92	2.76	9.19	27.6	91.9
1.8	0	0.27	0.91	2.73	9.1	27.3	91
2	0	0.27	0.9	2.7	9	27	90

2.2	0	0.27	0.89	2.67	8.91	26.7	89.1
2.4	0	0.26	0.88	2.64	8.81	26.4	88.1
2.6	0	0.26	0.87	2.62	8.72	26.2	87.2
2.8	0	0.26	0.86	2.59	8.63	25.9	86.3
3	0	0.26	0.85	2.56	8.54	25.6	85.4
3.2	0	0.25	0.84	2.53	8.45	25.3	84.5
3.4	0	0.25	0.84	2.51	8.36	25.1	83.6
3.6	0	0.25	0.83	2.48	8.27	24.8	82.7
3.8	0	0.25	0.82	2.46	8.19	24.6	81.9
4	0	0.24	0.81	2.43	8.1	24.3	81

Calibration settings

Calibration parameter settings for GUTS-RED-SD:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	1096	Log
mw	Yes	7.782E-5	99	Norm
hb	Yes	1E-6	0.07	Norm
bw	Yes	0.0005268	3339	Log
Fs	No	1	1	Norm

Calibration parameter settings for GUTS-RED-IT:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	2.865	Log
mw	Yes	7.782E-5	73.84	Log
hb	Yes	1E-6	0.07	Norm
bw	No	Inf	Inf	Norm
Fs	Yes	1.05	20	Log

Calibration results

Fitted parameters for GUTS-RED-SD:

Best fit parameter values and their 95% CI

kd: 0.001641 (0.001641* - 0.3428)
mw: 0.01201 (0.003358 - 2.312)
hb: 0.006361 (0.001094 - 0.01958)
bw: 5.76 (0.03123 - 8.508)

* edge of 95% parameter CI has run into a boundary
(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-SD):

Model efficiency (NSE, r-square): 0.6982

Normalised root-means-square error (NRMSE): 5.179 %

Minus log-likelihood (MLL): 84.37

AIC: 176.73

Survival probability prediction error (SPPE) for each treatment:

Data set	Treatment	Value
1	Control	0 %
1	T1	1.264 %
1	T2	-3.74 %
1	T3	1.264 %
1	T4	-7.08 %
1	T5	-21.7 %
1	T6	13.33 %

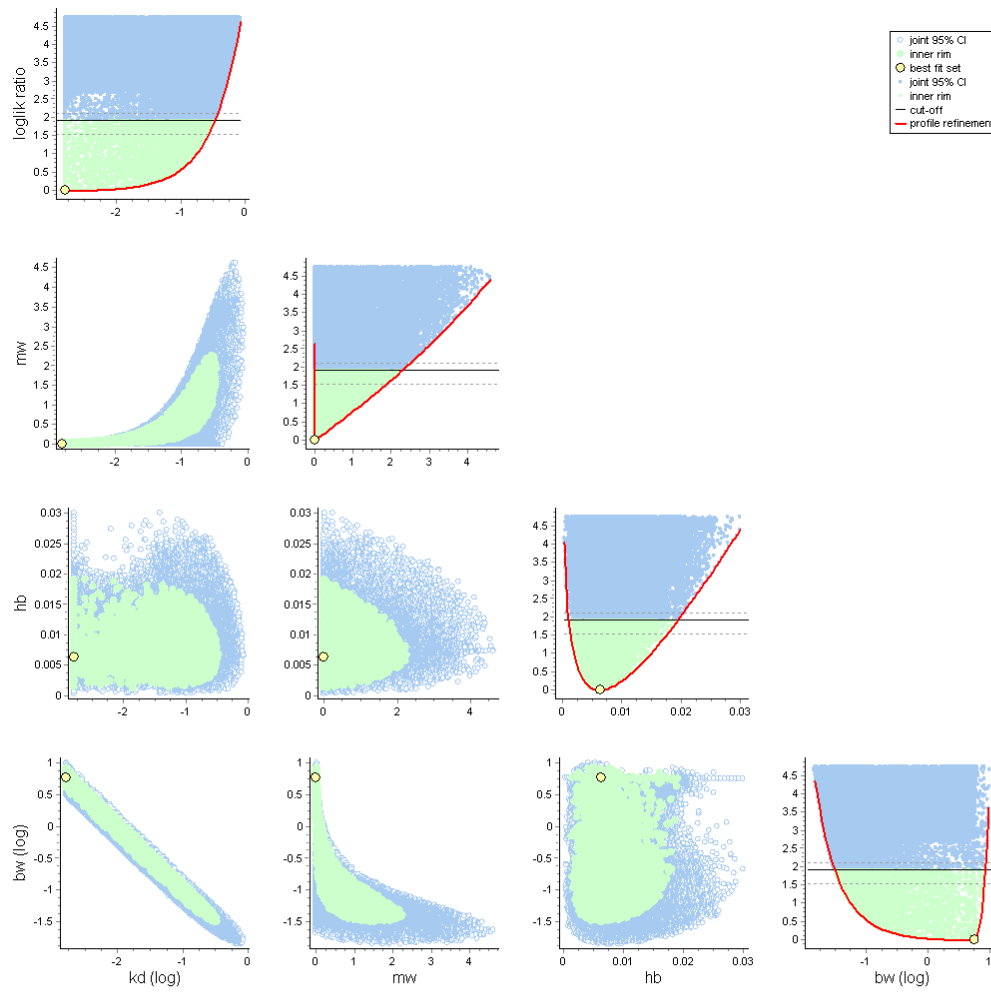
GUTS-RED-SD results table for LC_{x,t} [[C]], with 95% CI:

Time [d]	LC50	LC20	LC10
1	161 (117.2 - 233.9)	61 (45.91 - 83.04)	35.43 (24.89 - 47.6)
2	43.72 (33.08 - 60.98)	18.41 (13.06 - 24.78)	11.76 (7.156 - 16.32)
3	20.92 (15.95 - 28.58)	9.513 (6.28 - 12.94)	6.441 (3.349 - 9.272)

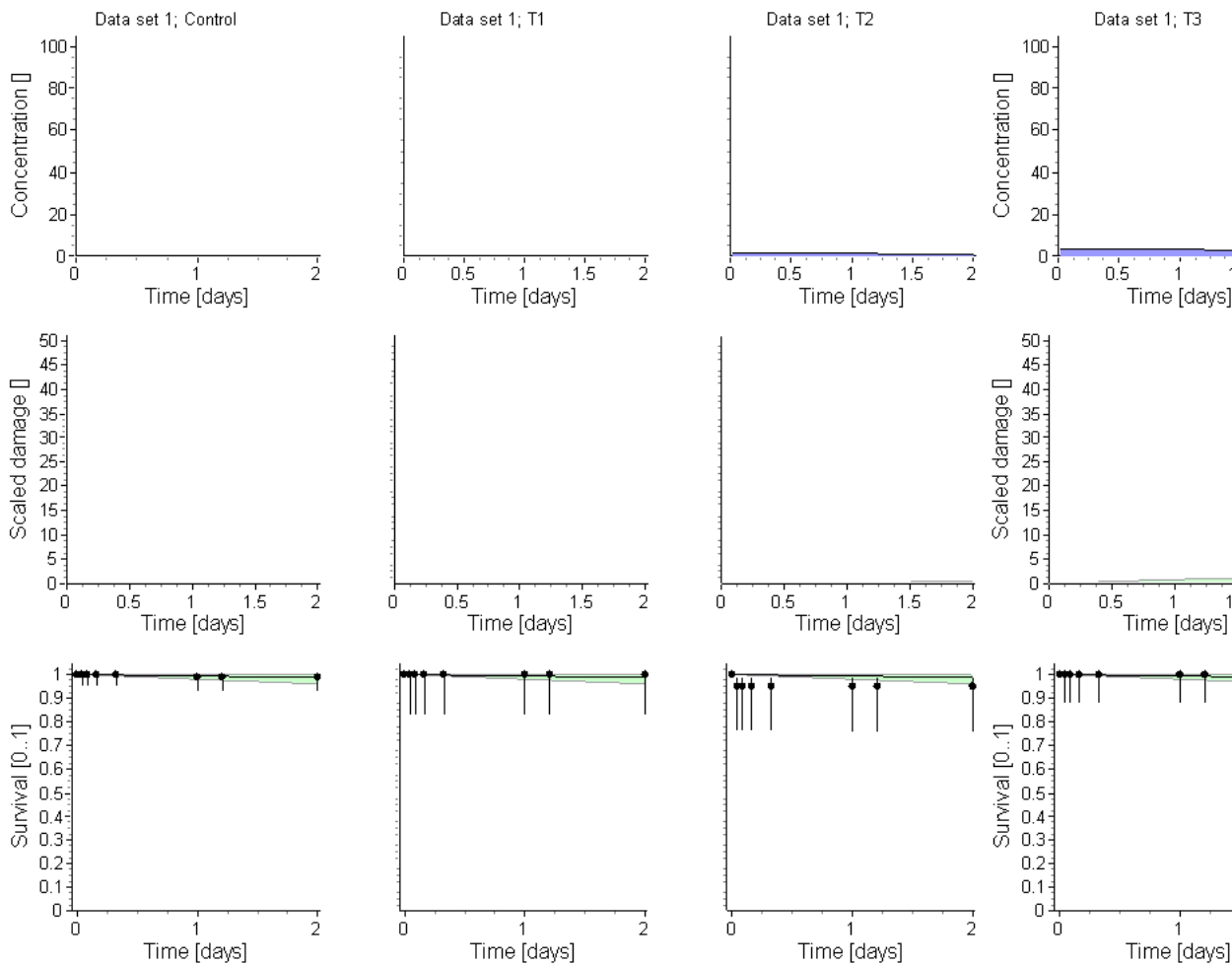
4	12.59 (9.461 - 17.52)	6.071 (3.761 - 8.739)	4.278 (1.977 - 6.566)
7	4.878 (3.402 - 8.445)	2.651 (1.326 - 5.01)	2.008 (0.7341 - 4.088)
10	2.75 (1.795 - 5.9)	1.611 (0.695 - 3.899)	1.27 (0.4015 - 3.336)
14	1.639 (0.975 - 4.605)	1.027 (0.3846 - 3.315)	0.8377 (0.2324 - 2.955)
21	0.9049 (0.4594 - 3.696)	0.6107 (0.1937 - 2.927)	0.5162 (0.1239 - 2.71)
27	0.6363 (0.2913 - 3.359)	0.4478 (0.1287 - 2.785)	0.3859 (0.08539 - 2.626)
28	0.6052 (0.2729 - 3.319)	0.4284 (0.1215 - 2.769)	0.3701 (0.081 - 2.617)
42	0.3524 (0.1341 - 2.984)	0.2647 (0.06482 - 2.636)	0.2349 (0.04579 - 2.543)
50	0.2818 (0.0997 - 2.885)	0.2166 (0.05005 - 2.599)	0.1941 (0.03619 - 2.523)
100	0.122 (0.03286 - 2.648)	0.1013 (0.01918 - 2.514)	0.09384 (0.01511 - 2.479)
182	0.06345 (0.01396 - 2.553)	0.05544 (0.009209 - 2.482)	0.05246 (0.007721 - 2.463)

Plots for GUTS-RED-SD calibration:

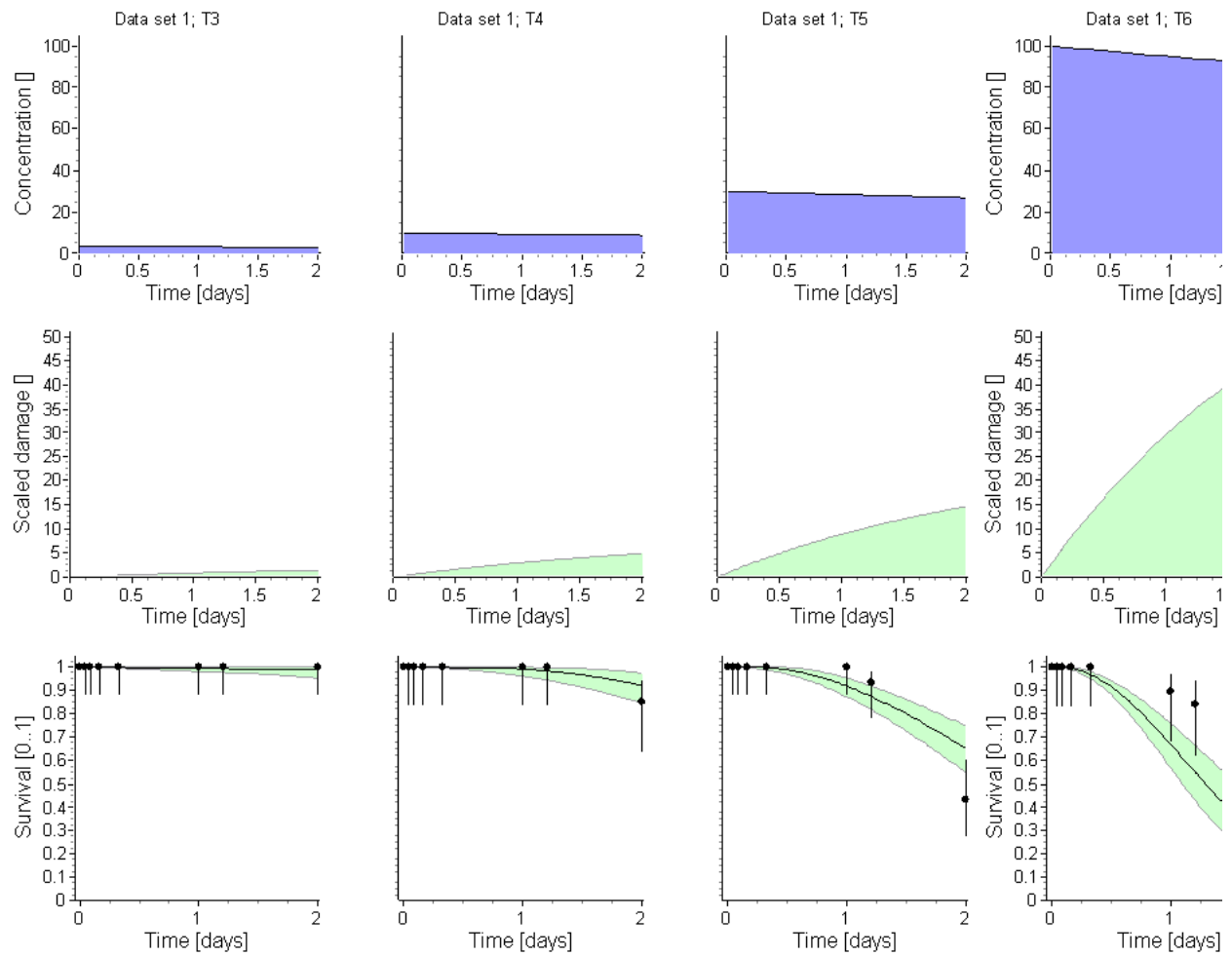
Parameter space plot for the calibration of GUTS-RED-SD:



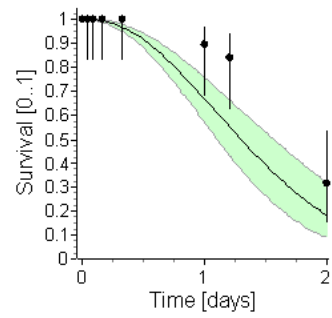
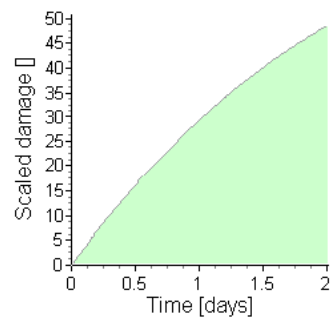
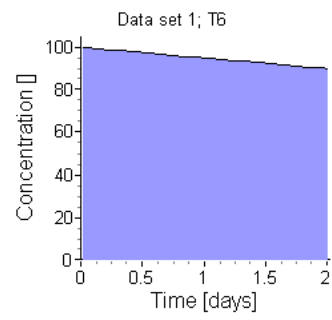
Exposure, damage and survival plots for the calibration of GUTS-RED-SD:



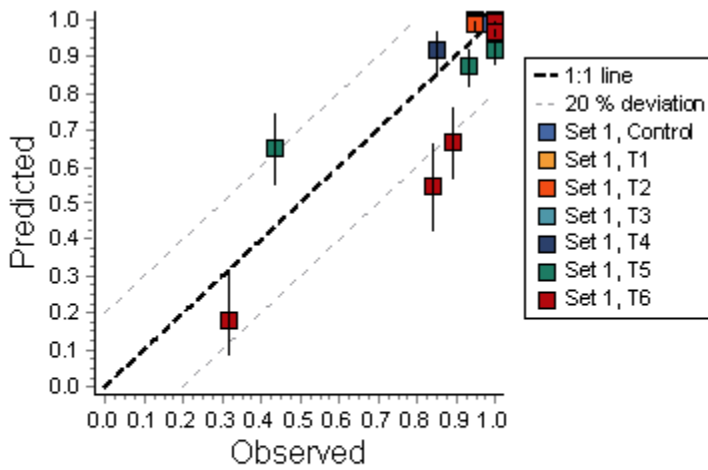
... continued plot:



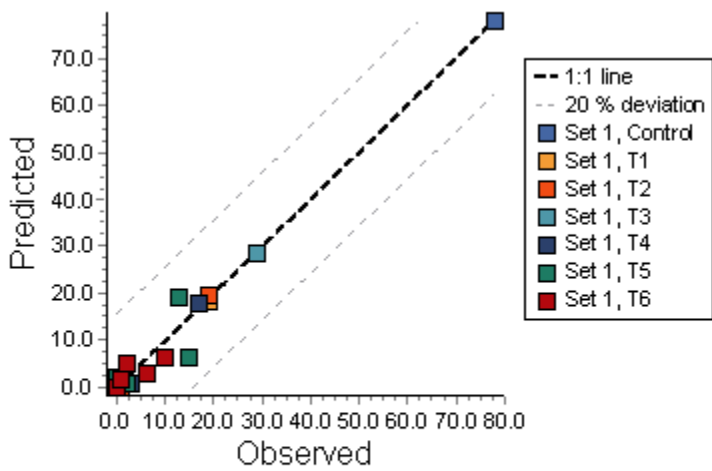
... continued plot:



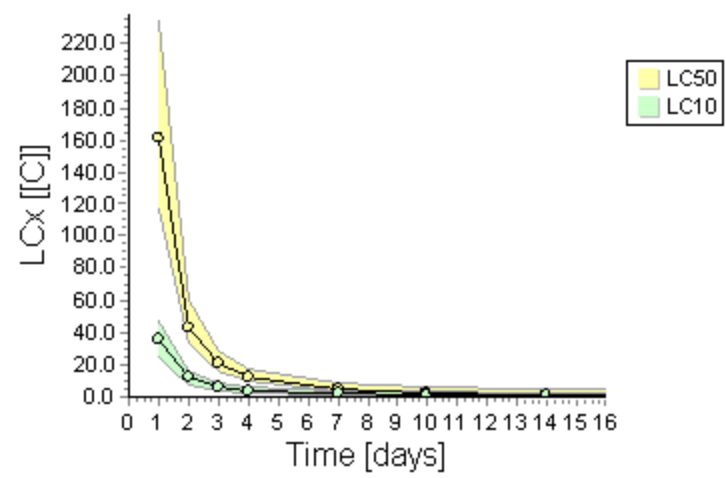
Observed vs. Predicted survival plot for the calibration of GUTS-RED-SD:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-SD:



LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-SD):



Fitted parameters for GUTS-RED-IT:

Best fit parameter values and their 95% CI

kd: 0.001641 (0.001641* - 0.09744)

mw: 0.1537 (0.1193 - 8.629)

hb: 0.008281 (0.001426 - 0.0247)

Fs: 5.828 (3.872 - 10.31)

* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-IT):

Model efficiency (NSE, r-square): 0.3093

Normalised root-means-square error (NRMSE): 7.539 %

Minus log-likelihood (MLL): 97.04

AIC: 202.08

Survival probability prediction error (SPPE) for each treatment:

Data set	Treatment	Value
1	Control	0.3767 %
1	T1	1.645 %
1	T2	-3.33 %
1	T3	1.933 %
1	T4	-9.93 %
1	T5	-29.2 %
1	T6	13.12 %

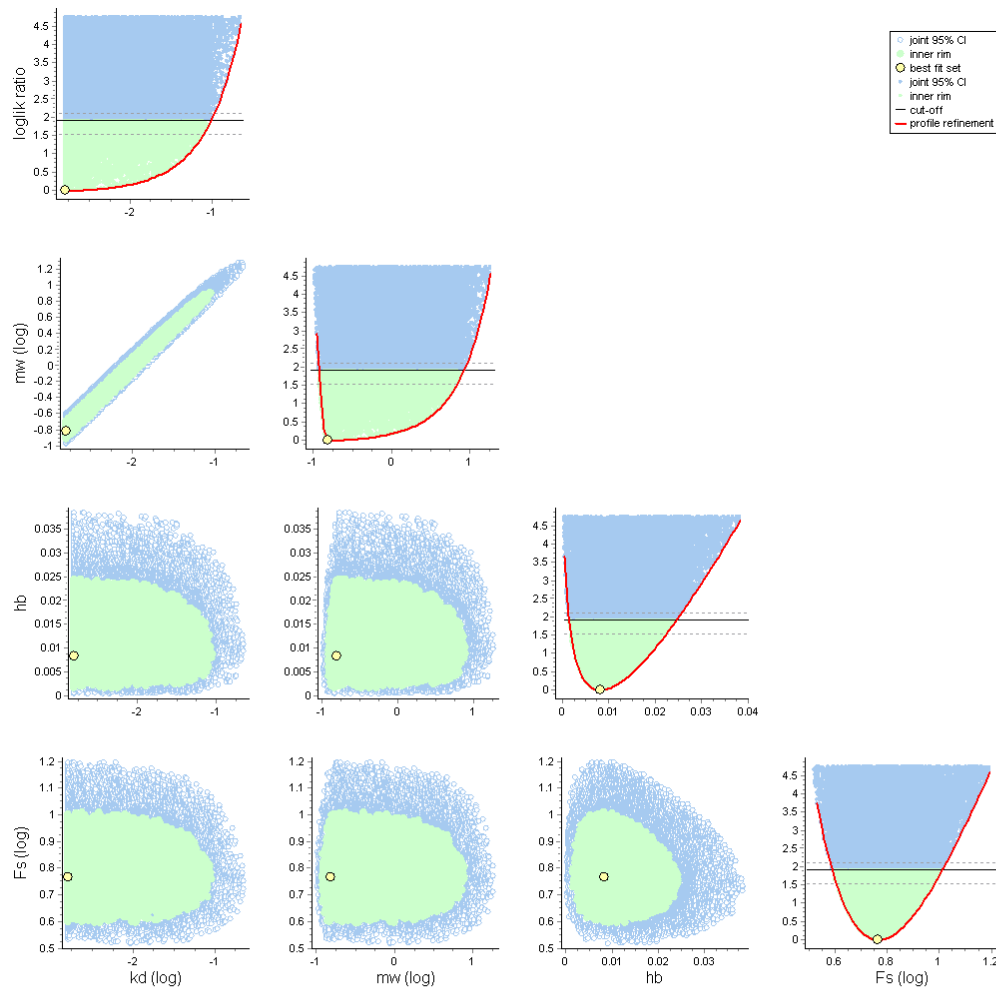
GUTS-RED-IT results table for LC_{x,t} [[C]], with 95% CI:

Time [d]	LC50	LC20	LC10
1	93.72 (71.52 - 127.5)	48.1 (35.4 - 64.92)	32.56 (22.04 - 46.08)
2	46.9 (35.79 - 63.83)	24.07 (17.74 - 32.49)	16.29 (11.03 - 23.06)
3	31.29 (23.88 - 42.59)	16.06 (11.85 - 21.68)	10.87 (7.363 - 15.39)
4	23.49 (17.92 - 31.97)	12.06 (8.903 - 16.28)	8.161 (5.528 - 11.55)
7	13.45 (10.27 - 19.03)	6.906 (5.114 - 9.829)	4.675 (3.169 - 6.902)

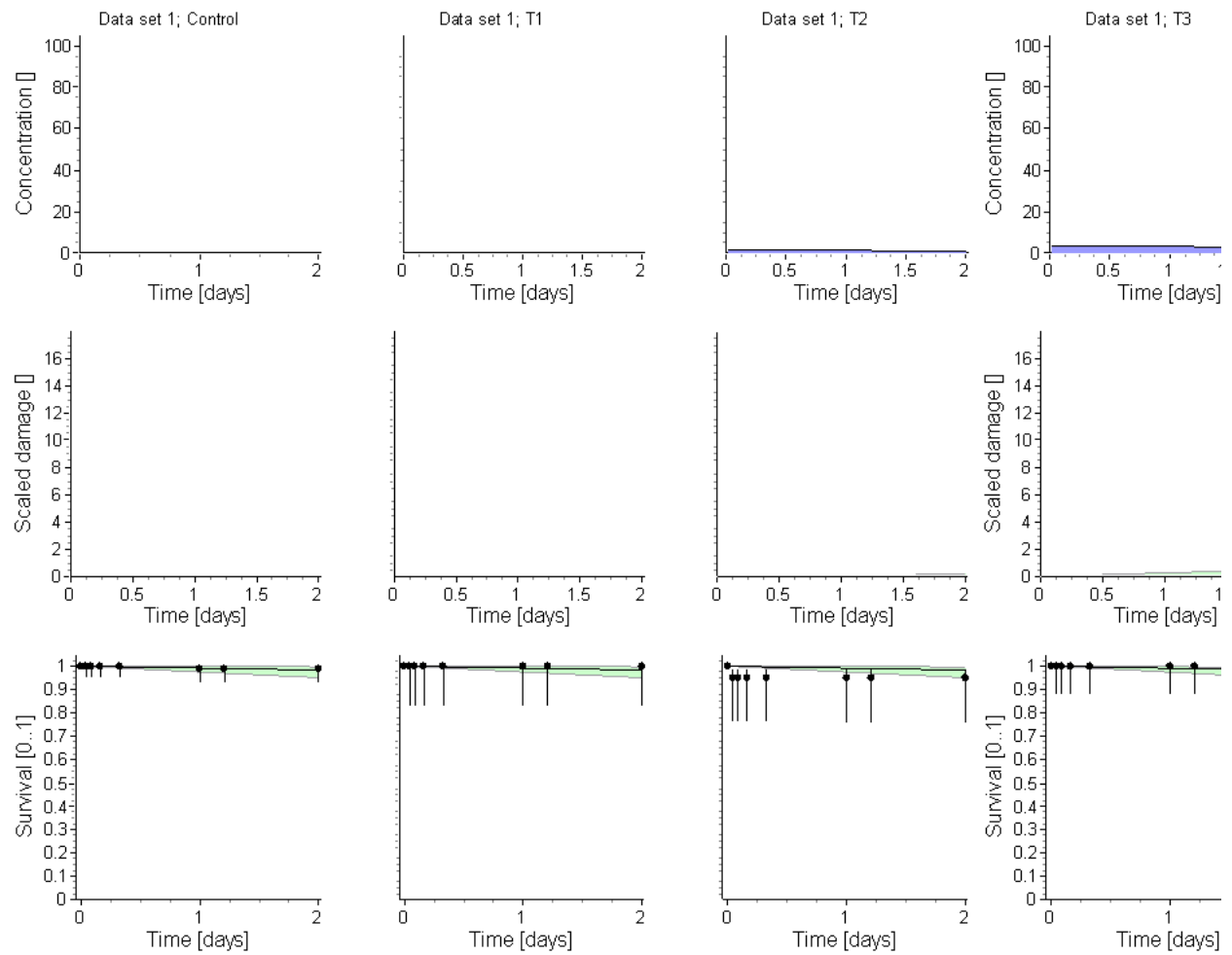
10	9.441 (7.205 - 14.64)	4.846 (3.592 - 7.651)	3.28 (2.225 - 5.359)
14	6.766 (5.163 - 12.19)	3.473 (2.578 - 6.251)	2.351 (1.596 - 4.378)
21	4.537 (3.462 - 10.39)	2.328 (1.729 - 5.238)	1.576 (1.072 - 3.623)
27	3.546 (2.706 - 9.726)	1.82 (1.351 - 4.902)	1.232 (0.8392 - 3.331)
28	3.422 (2.611 - 9.653)	1.756 (1.304 - 4.867)	1.189 (0.8101 - 3.298)
42	2.307 (1.761 - 9.151)	1.184 (0.8791 - 4.63)	0.8017 (0.5481 - 3.133)
50	1.951 (1.489 - 9.062)	1.001 (0.7433 - 4.589)	0.6778 (0.4642 - 3.105)
100	1.015 (0.7748 - 8.985)	0.5211 (0.3869 - 4.555)	0.3528 (0.2438 - 3.082)
182	0.5952 (0.4542 - 8.984)	0.3055 (0.2268 - 4.555)	0.2068 (0.1429 - 3.082)

Plots for GUTS-RED-IT calibration:

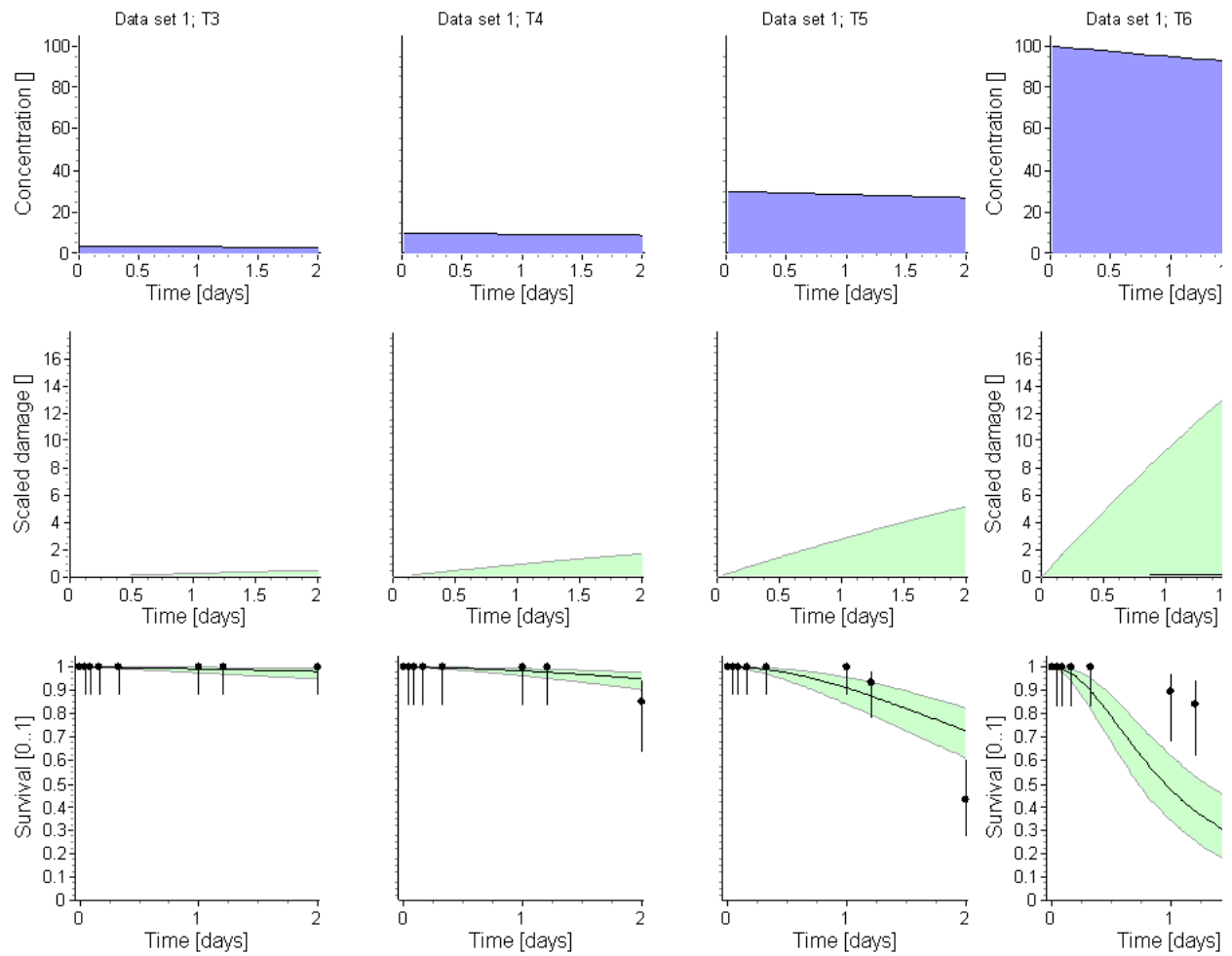
Parameter space plot for the calibration of GUTS-RED-IT:



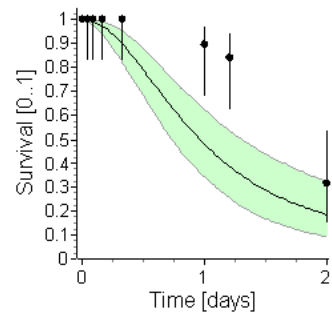
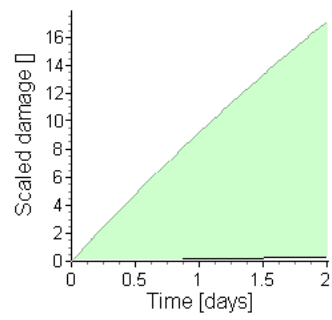
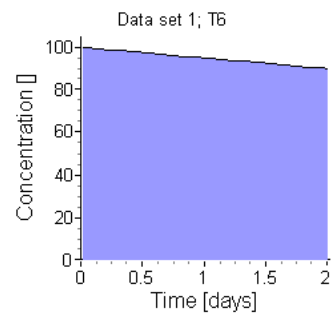
Exposure, damage and survival plots for the calibration of GUTS-RED-IT:



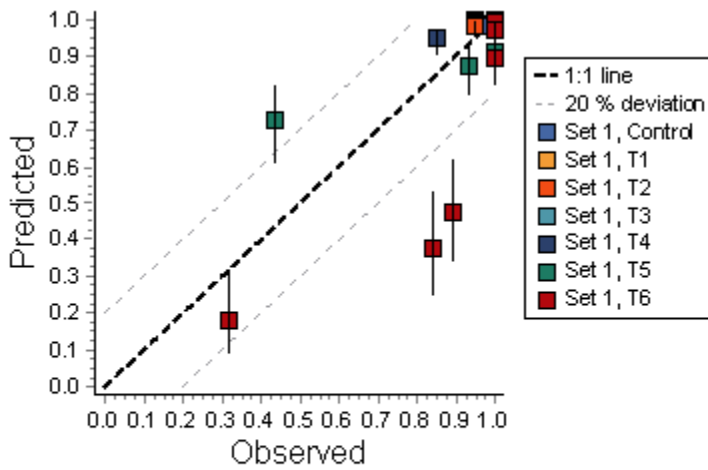
... continued plot:



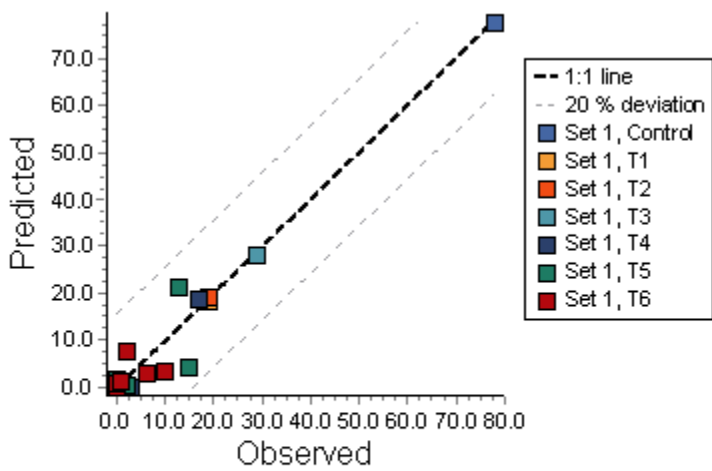
... continued plot:



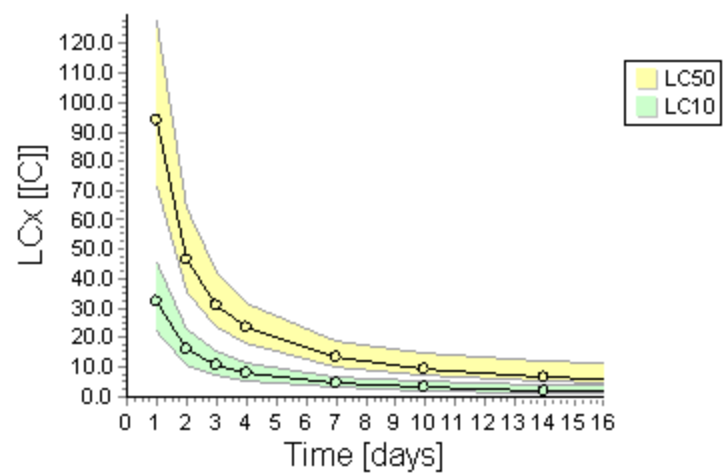
Observed vs. Predicted survival plot for the calibration of GUTS-RED-IT:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-IT:



LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-IT):



Validation

No validation performed!

Predictions

No predictions performed!